

ALF Citations

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Just a collection of some list of publications citing **ALF**:

ALF 1.0 Martin Bercx, Florian Goth, Johannes S. Hofmann, Fakher F. Assaad. “The ALF (Algorithms for Lattice Fermions) project release 1.0. Documentation for the auxiliary field quantum Monte Carlo code”. SciPost Phys. 3, 013 (2017).

ALF 2.0 Martin Bercx, Florian Goth, Johannes S. Hofmann, Jefferson S. E. Portela, Jonas Schwab, Fakher F. Assaad. “The ALF (Algorithms for Lattice Fermions) project release 2.0”. *Draft*

either for context (**Literature**), for reference data (**Benchmark**), or for obtaining results (**Main tool**).

Main sources: [Google Scholar](#), arxiv for arxiv entries and university repositories for thesis and dissertations.

How ALF 1.0 is cited

Reference	Title	ALF role
[1] Sato, Hohenadler, and Assaad (2017)	“Dirac fermions with competing orders: Non-Landau transition with emergent symmetry”	Main tool
[2] Gazit et al. (2018)	“Confinement transition of $\mathbb{Z}_2$ gauge theories coupled to massless fermions: Emergent quantum chromodynamics and SO(5) symmetry”	Main tool
[3] Tang et al. (2018)	“The role of electron-electron interactions in two-dimensional Dirac fermions”	Main tool
[4] Broecker, Assaad, and Trebst (2017)	<i>Quantum phase recognition via unsupervised machine learning</i>	Main tool
[5] Beyl, Goth, and Assaad (2018)	“Revisiting the hybrid quantum Monte Carlo method for Hubbard and electron-phonon models”	Benchmark
[6] Huffman and Chandrasekharan (2017)	“Fermion bag approach to Hamiltonian lattice field theories in continuous time”	Literature
[7] Chen et al. (2019)	“Charge-density-wave transitions of Dirac fermions coupled to phonons”	Literature
[8] Toldin and Assaad (2018)	“Entanglement Hamiltonian of interacting fermionic models”	Main tool
[9] Sato, Assaad, and Grover (2018)	“Quantum Monte Carlo simulation of frustrated Kondo lattice models”	Main tool
[10] Liu et al. (2019)	“Superconductivity from the condensation of topological defects in a quantum spin-Hall insulator”	Main tool
[11] Wang, Assaad, and Toldin (2017)	“Finite-size effects in canonical and grand-canonical quantum Monte Carlo simulations for fermions”	Main tool
[12] Ippoliti et al. (2018)	“Half-filled Landau levels: a continuum and sign-free regularization for three-dimensional quantum critical points”	Main tool
[13] Hohenadler and Assaad (2018)	“Fractionalized metal in a Falicov-Kimball model”	Main tool

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[14] Fukuma, Matsumoto, and Umeda (2019)	“Applying the tempered Lefschetz thimble method to the Hubbard model away from half filling”	Benchmark
[15] Raczkowski and Assaad (2019)	“Emergent coherent lattice behavior in Kondo nanosystems”	Main tool
[16] Shinaoka et al. (2017)	“Quantum Monte Carlo impurity solvers for multi-orbital problems and frequency-dependent interactions”	Main tool
[17] Ulybyshev, Winterowd, and Zafeiropoulos (2020)	“Lefschetz thimbles decomposition for the Hubbard model on the hexagonal lattice”	Benchmark
[18] Wynen et al. (2019)	“Avoiding ergodicity problems in lattice discretizations of the Hubbard model”	Literature
[19] Kim et al. (2019)	“Dimensional crossover in the SU (4) Heisenberg model in the six-dimensional antisymmetric self-conjugate representation revealed by quantum Monte Carlo and linear flavor-wave theory”	Literature
[20] Buividovich et al. (2019)	“Numerical evidence of conformal phase transition in graphene with long-range interactions”	Literature
[21] Raczkowski et al. (2020)	“Hubbard model on the honeycomb lattice: From static and dynamical mean-field theories to lattice quantum Monte Carlo simulations”	Main tool
[22] Hofmann, Assaad, and Grover (2019)	“Fractionalized Fermi liquid in a frustrated Kondo lattice model”	Main tool
[23] Huffman and Chandrasekharan (2020)	“Fermion-bag inspired Hamiltonian lattice field theory for fermionic quantum criticality”	Benchmark
[24] Toldin, Sato, and Assaad (2019)	“Mutual information in heavy-fermion systems”	Main tool
[25] Leaw et al. (2019)	“Electronic ground state in bilayer graphene with realistic Coulomb interactions”	Main tool
[26] Toldin and Assaad (2019)	“Entanglement studies of interacting fermionic models”	Main tool
[27] Peri et al. (2020)	<i>Fragile topology and flat-band superconductivity in the strong-coupling regime</i>	Main tool
[28] Raczkowski and Assaad (2020)	“Phase diagram and dynamics of the SU (N) symmetric Kondo lattice model”	Main tool
[29] Hofmann, Berg, and Chowdhury (2020)	<i>Superconductivity, pseudogap, and phase separation in topological flat bands: a quantum Monte Carlo study</i>	Main tool
[30] Gazit, Assaad, and Sachdev (2020)	<i>Fermi-surface reconstruction without symmetry breaking</i>	Literature
[31] Hohenadler and Assaad (2019)	“Orthogonal metal in the Hubbard model with liberated slave spins”	Main tool
[32] Danu, Assaad, and Mila (2019)	“Exploring the Kondo effect of an extended impurity with chains of Co adatoms in a magnetic field”	Main tool
[33] Sato et al. (2020)	<i>Topological terms on topological defects: a quantum Monte Carlo study</i>	Main tool
[34] Danu et al. (2020)	<i>Kondo breakdown in a spin-1/2 chain of adatoms on a Dirac semimetal</i>	Main tool
[35] Grossman et al. (2020)	<i>Specific Heat of a Quantum Critical Metal</i>	Main tool
[36] Kappl et al. (2020)	“Statistical error estimates in dynamical mean-field theory and extensions thereof”	Literature
[37] Huffman (2019)	“Fermion Bag Approach for Hamiltonian Lattice Field Theories”	Benchmark
[38] Wynen (2020)	“Strongly interacting few-body systems from lattice stochastic methods”	Literature
[39] Zhang et al. (2020)	“Coexistence and interaction of spinons and magnons in an antiferromagnet with alternating antiferromagnetic and ferromagnetic quantum spin chains”	Main tool

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[40] Assaad (2018)	“Quantum Monte Carlo Methods for Fermion-Boson Problems”	Main tool
[41] Breoni (2019)	“Torus spectroscopy of the chiral Heisenberg quantum phase transition with quantum Monte Carlo simulations”	Main tool
[42] Hofmann (2020)	“On the interplay of topology and interaction: A quantum Monte Carlo study”	Main tool
[43] Hofmann et al. (2020)	“Search for correlation-induced adiabatic paths between distinct topological insulators”	Main tool
[44] Bröcker (2018)	“Disentangling and machine learning the many-fermion problem”	Main tool
[45] Wang et al. (2020)	<i>Doping-induced quantum spin Hall insulator to superconductor transition</i>	Main tool
[46] Beyl (2020)	“Hybrid Quantum Monte Carlo for Condensed Matter Models”	Main tool
[47] Rammelmüller (2020)	“Exploring imbalanced Fermi gases with stochastic quantization”	Literature

Citations with no ALF Team member participation

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[6] Huffman and Chandrasekharan (2017)	“Fermion bag approach to Hamiltonian lattice field theories in continuous time”	Literature
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[18] Wynen et al. (2019)	“Avoiding ergodicity problems in lattice discretizations of the Hubbard model”	Literature
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Citations from *papers* with no ALF Team member participation

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[36] Kappl et al. (2020)	“Statistical error estimates in dynamical mean-field theory and extensions thereof”	Literature

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- [3] Ho-Kin Tang et al. “The role of electron-electron interactions in two-dimensional Dirac fermions”. In: *Science* 361.6402 (2018), pp. 570–574.
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- [5] Stefan Beyl, Florian Goth, and Fakher F Assaad. “Revisiting the hybrid quantum Monte Carlo method for Hubbard and electron-phonon models”. In: *Physical Review B* 97.8 (2018), p. 085144.
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- [19] Francisco H Kim et al. “Dimensional crossover in the SU (4) Heisenberg model in the six-dimensional anti-symmetric self-conjugate representation revealed by quantum Monte Carlo and linear flavor-wave theory”. In: *Physical Review B* 100.8 (2019), p. 085103.
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